

GLOBAL LOGISTICS & DEMAND FORECASTING

Project Overview:

Build a scalable production solution that can seamlessly process, analyse, and forecast massive datasets moving well beyond the limits of local computer memory.

Key Analytics Techniques:

Basic Core Metrics: Real-time tracking of active delivery statuses and current warehouse inventory levels.

Route Efficiency Analysis: Calculating and visualizing massive-scale variances between Actual Time of Arrival (ATA) and Estimated Time of Arrival (ETA) at an individual shipment level.

Demand Forecasting: Implementing moving average logic and more advanced time-series methods applied to millions of distinct Stock Keeping Units (SKUs) for massive dataset

Weekly Development Plan:

Week 1 - Environment & Ingestion

Planning - setup the local spark using optimised py spark data frames

Week 2 - Big Data Processing

Perform the necessary logiscale aggregations within spark over massive data set

Week 3 -Visualization Layer - Exporting the final Aggregations dataset to Power BI

Week 4 -Final Delivery - Creating Control Tower Dashboard View To get Scalability ,Performance ,Resilience of the Big Data solution

Technology Stack:

primary Data Processing: PySpark

Storage Layer: SQL Database

Visualization Layer: Power BI

Deployment/Environment: Docker or Local Standalone mode for setting up Spark.

Business Impact:

System Stability

Operational Efficiency

Cost Reduction

📌 Project Outcome:

The Logistic-Scale provides a complete analytics ,Scalability ,Performance ,Resilience from raw sales data to business insights using SQL, Python, and Power BI dashboards.

Final Dashboard:

